THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **How do Organisms Reproduce?** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Asexual reproduction produces offspring that are not exactly identical to the parent. Reason (R): Copying of DNA is accompanied by small errors. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A plant breeder crosses two varieties and also grows a plant by stem cutting. The offspring from the cross show 1

(a) variation in the cutting only
(b) no variation in either
(c) identical traits in the cross
(d) variation, and those from the cutting are identical to the parent

3. An organism with a very complex body that can still regenerate would need 1

(a) no DNA
(b) undifferentiated cells to produce missing parts
(c) no cells at all
(d) only gametes

4. Assertion (A): Regeneration is the same as reproduction in all organisms. Reason (R): Planaria grows from a cut piece. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Rhizopus is grown on bread in a sealed, completely dry, sterile packet. Growth is absent because 1

(a) sporangia are absent **(b)** spores need moisture to germinate
(c) spores are not produced **(d)** bread kills them

6. If the placenta fails to form properly, the main problem for the embryo is 1

(a) excess heat **(b)** lack of DNA
(c) lack of chromosomes **(d)** shortage of nutrients and oxygen

7. Assertion (A): Testes of mammals lie outside the abdominal cavity. Reason (R): Sperm formation requires a lower temperature than the body temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A single Amoeba undergoes binary fission every 20 minutes. Starting with one Amoeba, the number after 1 hour is 1

(a) 6 **(b)** 16 **(c)** 3 **(d)** 8

9. **Assertion (A):** Plants raised by vegetative propagation show characters identical to the parent plant. 1
Reason (R): Vegetative propagation involves only one parent and no fusion of gametes.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Hydra reproduces by budding. 1
Reason (R): Planaria can regenerate a complete individual from a cut piece.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A tiny part of a Hydra body is used to grow a new Hydra in a lab. Is this regeneration or budding? Justify. 2

12. In which situation would a plant need a pollinator the most: a unisexual flower or a bisexual flower? Give a reason. 2

13. Name the structure that develops into the male gamete carrier in flowering plants and state where it is formed. 2

14. Starting with one Amoeba and dividing every 30 minutes, find the number after 3 hours. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. After 2 hours, the number of Amoebae from one cell that divides every 15 minutes is calculated. Find it and state the type of reproduction. 3

16. Why does cutting a plant stem and keeping it in moist soil give a new plant while cutting a human finger does not give a new human? 3

17. What would happen if the uterine lining did not thicken each month? 3

OR

Explain why a woman who has her ovaries removed will not have menstrual cycles. 3

18. A flower has a long style. What is the effect on the pollen tube and how does it reach the ovule? 3

OR

What are the advantages and disadvantages of spore formation for a fungus? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Lakshmi dissected a hibiscus flower in her biology class. She saw sepals, petals, several stamens and a pistil in the centre. A bee had been visiting the flower and carrying yellow dust from one flower to another. 5
(i) Name the male reproductive part of the flower.
(ii) What is the transfer of pollen from anther to stigma called?
(iii) After fertilisation, which part of the flower develops into the fruit?
(iv) Differentiate between self-pollination and cross-pollination and name one agent of cross-pollination.

20. A student sets up three petri dishes of bread: one moist and warm, one dry, one moist and cold. Predict the results and explain with respect to spores. 5

OR

Justify: 5
(a) a bean seed contains stored food
(b) pollen grains are produced in large numbers
(c) egg is larger than sperm.

END OF QUESTION PAPER — Total: 40 Marks